20 Show that $\dfrac{12+\sqrt{128}}{1-\sqrt{2}}$ can be written in the form $a + b\sqrt{2}$ where a and b are integers.

$\sqrt{128} = \sqrt{(64 \times 2)} = 8\sqrt{2}$

1 mark
Simplify surd.

$$\frac{12 + 8\sqrt{2}}{1 - \sqrt{2}} \times \frac{1 + \sqrt{2}}{1 + \sqrt{2}}$$

1 mark
Multiply by conjugate.

	1	$\sqrt{2}$
12	12	$12\sqrt{2}$
$8\sqrt{2}$	$8\sqrt{2}$	16

Numerator = $28 + 20\sqrt{2}$

1 mark
Expand numerator.

Denominator = $(1 - \sqrt{2})(1 + \sqrt{2}) = 1 - 2 = -1$

$$\frac{28 + 20\sqrt{2}}{-1}$$

$= -28 - 20\sqrt{2}$

1 mark
a = −28,
b = −20.

(Total for Question 20 is 4 marks)